

Brute Force Detection and Automated Response – Architecture and Workflow

1. Use Case Overview

This use case demonstrates the detection and automated response of brute force login attacks using Splunk Enterprise and Splunk SOAR.

The solution monitors Windows Security logs for multiple failed login attempts. When the number of failed logins exceeds a predefined threshold, Splunk generates an alert and forwards the event to Splunk SOAR. Splunk SOAR then executes an automated response playbook that disables the targeted user account in Active Directory.

2. Business Objective

Brute force attacks are commonly used by attackers to gain unauthorized access to systems through repeated authentication attempts.

The objective of this use case is to:

- Detect repeated failed login attempts.
- Reduce analyst response time.
- Automate containment actions.
- Demonstrate end-to-end SOC automation capabilities.

3. Solution Components

Data Source

- Windows Server 2022
- Windows Client

Log Collection

- Splunk Universal Forwarder
- Splunk Heavy Forwarder

SIEM Platform

- Splunk Enterprise Search Head
- Splunk Enterprise Indexer

SOAR Platform

- Splunk SOAR 7.1.0.225

Response Integration

- Microsoft AD LDAP Application

4. Architecture Diagram

Diagram 1: High-Level Architecture

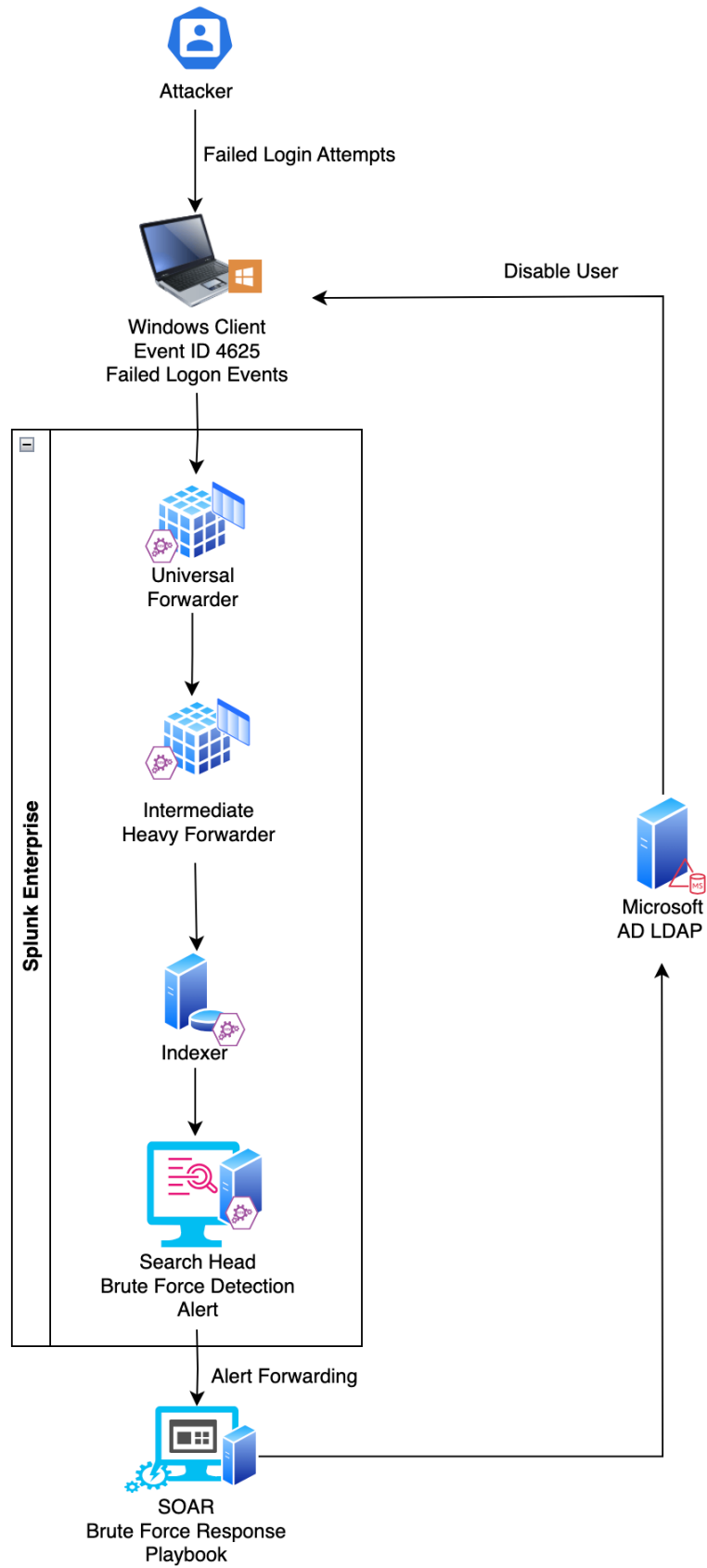
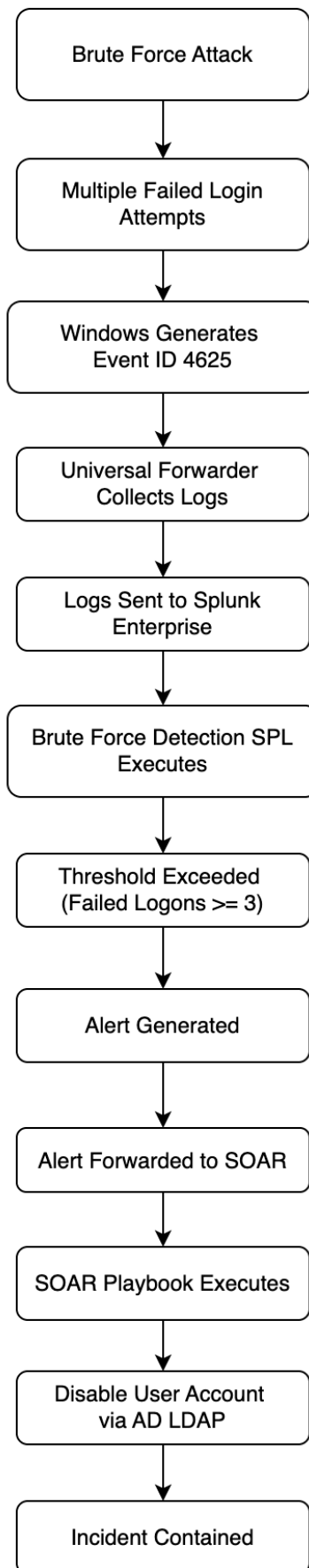


Diagram 2: Detection & Response Workflow



5. Detection Workflow

Step 1 – Attack Execution

An attacker performs multiple failed login attempts against a Windows user account.

Step 2 – Event Generation

Windows records failed authentication attempts as Event ID 4625.

Step 3 – Log Collection

Splunk Universal Forwarder collects the events and forwards them through the Heavy Forwarder into Splunk Enterprise.

Step 4 – Detection

A scheduled Splunk search identifies accounts that exceed the configured failed login threshold.

Step 5 – Alert Generation

Splunk generates a brute force detection alert.

Step 6 – Event Forwarding

The alert is forwarded to Splunk SOAR.

Step 7 – Automated Response

Splunk SOAR executes a playbook that disables the affected account using Microsoft Active Directory LDAP integration.

Step 8 – Incident Containment

The account is disabled, preventing further login attempts and limiting the impact of the attack.

6. Security Benefits

- Early detection of password guessing attacks.
- Automated response without analyst intervention.
- Reduced mean time to respond (MTTR).
- Improved SOC operational efficiency.
- Enhanced protection of Active Directory accounts.

7. Expected Outcome

Upon detection of a brute force attack:

- Splunk identifies the malicious activity.
- Splunk SOAR receives the alert.
- The playbook executes automatically.
- The user account is disabled.
- The attack is contained.

Next Steps

Creating Alerts and Triggering them from Splunk Enterprise to SOAR – Refer **17_Creating_Bruteforce_alert_on_Splunk_Enterprise_V1.pdf**.